

KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

Mathematics
Year 8

EXTENSION WORKBOOK | E1–E2 SELF-STUDY

Probability

Year 8 Mathematics • Extension workbook

Name _____ Class _____ Teacher _____

How this workbook works

Each lesson is **three pages**. Copy your teacher's **notes**, then work through **We do** (dark boxes: we solve these together; copy every step) and **You do** (your turn, one round at a time). Rate each success criterion at the end.

Your notes and examples match the main workbook, so you follow the class lesson, then do these harder questions: page 1 **Medium** ●●, page 2 **Hard** ●●●, page 3 **Going further**. E1 and E2 are **self-study**: their key ideas are printed and their examples are worked for you.

NSW Mathematics K–10 Syllabus (2022), Stage 4: MA4-PRO-C-01; extension (Stage 5 Path): MA5-PRO-P-01. We acknowledge the Bundjalung people, the Traditional Custodians of the land on which we learn.

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Cover and banner artwork: *Kingscliff High School Vision Statement* by local Indigenous artist Marni Tuala.

1 Chance and the sample space

Probability • MA4-PRO-C-01 **MEDIUM ••**

LEARNING INTENTION

We are learning to describe chance with words and with numbers from 0 to 1, list a sample space, and decide whether outcomes are equally likely.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* place an event on the probability scale from 0 to 1. ○○○
- *I can* list the sample space of a chance experiment. ○○○
- *I can* decide, with a reason, whether outcomes are equally likely. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Answer each part. Give a reason where you are asked to.

a The scale

Mark: A the sun rises tomorrow; B rolling a 7 on a die; C a coin lands heads.



b Sample space

List the sample space. How many outcomes are there?



c Equally likely?

Is each colour equally likely? Explain.



d Sample space

List the sample space for the day of the week a person was born on. How many outcomes?

YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER impossible = 0 · unlikely: between 0 and $\frac{1}{2}$ · even chance = $\frac{1}{2}$ · likely: between $\frac{1}{2}$ and 1 · certain = 1

ROUND 1 Questions 1–4: count the blue sectors, then describe the chance of landing on blue in words.

1



Blue / total _____

Word _____

2



Blue / total _____

Word _____

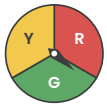
3



Blue / total _____

Word _____

4



Blue / total _____

Word _____

ROUND 2 Questions 5–8: list the sample space and count the outcomes. Watch out for repeats.

5 Spinning this spinner



Sample space { _____ }

Outcomes _____

6 Choosing one tile



Sample space { _____ }

Outcomes _____

7 The last digit of a car number plate

Sample space { _____ }

Outcomes _____

8 The number of heads when 3 coins are tossed together



Sample space { _____ }

Outcomes _____

1 Chance and the sample space

Probability • MA4-PRO-C-01

HARD •••

YOU DO on your own: keep going, one round at a time. Show your working. **12 QUESTIONS**

REMEMBER sample space: each outcome once · event: the outcomes we want · compare events by counting their outcomes · a probability is never below 0 or above 1

ROUND 3 Questions 9–11: list the outcomes in each event, in braces { }.

9 A 10-sided die numbered 1 to 10 shows a prime number.



Event { }

10 A month with exactly 30 days is chosen.

Event { }

11 A normal die shows a factor of 12.



Event { }

ROUND 4 Questions 12–14: are the outcomes equally likely? Write *yes* or *no*, then give a reason.

12 The number of heads when 3 coins are tossed.



Yes / no

Reason

13 The first letter of a person's first name.

Yes / no

Reason

14 A whole number from 1 to 20 made by a random number generator.



Yes / no

Reason

ROUND 5 Questions 15–17: order the events from least to most likely. Explain using the number of outcomes.

15 A die is rolled. A: an even number. B: 5 or more. C: less than 2.



Order

Reason

16 One tile is chosen. Event (i): a B. Event (ii): an I. Event (iii): a vowel.

Order

Reason

17 The spinner is spun. A: red. B: blue. C: yellow.



Order

Reason

ROUND 6 Questions 18–20: probabilities as numbers. Answer each one and show how you know.

18 Write a probability, as a decimal, for an event that is unlikely but not impossible.

Answer

19 Which is more likely: an event with probability 0.45, or one with probability $\frac{4}{9}$?

Answer

20 Explain why a probability can never be 1.2 or −0.3.

Answer

1 Chance and the sample space

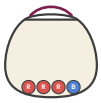
Probability • MA4-PRO-C-01

GOING FURTHER

WE DO with your teacher: copy every step into the space

Example 2 Answer in a full sentence, and give the reason.

a Equally likely?
One counter is taken. List the sample space. Are the outcomes equally likely?



b A match
Kingscliff plays Cabarita in a final. List the sample space. Are the outcomes equally likely?

c Order
Least to most likely: A an even number, B a 6, C less than 6, D a 9.



d Forecast
A 30% chance of rain: is rain likely or unlikely? Could it still rain?



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. 6 QUESTIONS

REMEMBER equally likely means the same chance, not just the same number of names · likely: more than half · unlikely: less than half, but not 0

1 Colour these 5 equal sectors so that red is twice as likely as blue, and green is impossible.



Answer

2 Spinner A has 3 equal sectors, 1 blue. Spinner B has 8 equal sectors, 3 blue. Which gives the better chance of blue? Explain.

Answer

3 A bag holds 10 red and blue counters. Red is likely; blue is unlikely but possible. How many red counters could there be? List every possibility.

Answer

4 Jo writes the sample space for this spinner as {red, red, blue}. What is wrong? Write it correctly.



Answer

5 Year 7
Order from least to most likely: A 0.6, B $\frac{3}{5}$, C 62%, D $\frac{5}{8}$.

Answer

6 A die is weighted so that a 6 comes up half the time. Are the six outcomes equally likely? Describe the chance of a 6.



Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. 3 PROBLEMS

1 The surf-club roster picks 1 of its 12 members at random: 5 are in Year 8, 4 in Year 9 and 3 in Year 10. Describe the chance that the person is in Year 8, and that they are in Year 11.

Answer

2 A spinner has 8 equal sectors coloured red, blue and yellow. Red is likely, blue and yellow are equally likely, and yellow is possible. Find every possible design.

Answer

3 Roll a die: Ava wins on 1 or 2, Ben on 3, 4 or 5, and Cal on 6. Is the game fair? Change the rules to make it fair.

Answer

2 Probability of an event

Probability • MA4-PRO-C-01

MEDIUM ••

LEARNING INTENTION

We are learning to find the probability of an event when the outcomes are equally likely, and write it as a fraction, decimal and percentage.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* count the favourable outcomes and the total outcomes. ○○○
- *I can* use P(E) = favourable ÷ total, and simplify. ○○○
- *I can* write a probability as a fraction, a decimal and a percentage. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Find each probability. Write the fraction in simplest form.

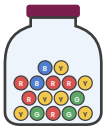
a Spinner: P(red)



b Die: P(an odd number)



c Lollies: P(yellow or blue)



d Tiles: P(a vowel)

K	I	N	G	S	C	L	I	F	F

YOU DO on your own: work down the box, one round at a time. Show your working. **8 QUESTIONS**

REMEMBER $P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$ · simplify: divide the top and bottom by the same number · a die: {1, 2, 3, 4, 5, 6}

ROUND 1 Questions 1–4: find P(blue) in simplest form. For unequal sectors, compare the sizes.

1



Working _____

P(blue) _____

2



Working _____

P(blue) _____

3



Working _____

P(blue) _____

4



Working _____

P(blue) _____

ROUND 2 Questions 5–8: a 10-sided die numbered 1 to 10 is rolled. Find each probability as a decimal.

5 P(a prime number)



Favourable { } _____

P = _____

6 P(a multiple of 3)



Favourable { } _____

P = _____

7 P(a factor of 10)



Favourable { } _____

P = _____

8 P(at least 8)



Favourable { } _____

P = _____

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER ‘or’ with no overlap: add the counts · decimal: top ÷ bottom · percentage: × 100% · missing total = count ÷ probability

ROUND 3 — Questions 9–11: one card is taken at random from a standard deck. Find each probability in simplest form.



J, Q, K are picture cards (shaded). An ace is not a number card here.

9 P(a black picture card)

Working

P =

10 P(a card with an even number)

Working

P =

11 P(a card from 2 to 5, inclusive)

Working

P =

ROUND 4 — Questions 12–14: one coin falls out of the money box from Example 2 at random. Give each answer in the form asked.



20 coins in the money box.

12 P(a 10c or 20c coin), as a percentage

Working

Answer

13 P(a coin worth less than $1), as a percentage

Working

Answer

14 P(a coin worth at least 50c), as a decimal

Working

Answer

ROUND 5 — Questions 15–17: find the missing number. Show your working.

15 A bag has 6 red counters and some blue ones. P(red) $= \frac{1}{3}$. How many blue counters?

Answer

16 A spinner has 12 equal sectors. P(green) $= 0.25$. How many sectors are green?

Answer

17 8K has 25 students. For a student chosen at random, P(plays a sport) $= \frac{3}{5}$. How many play a sport?

Answer

ROUND 6 — Questions 18–20: write each probability as a fraction, a decimal and a percentage. Round decimals to 2 places if needed.

18 One tile from P R O B A B I L I T Y: P(a vowel)

Fraction

Decimal

Percentage

19 A whole number from 1 to 40: P(a multiple of 8)

Fraction

Decimal

Percentage

20 A month chosen at random: P(it starts with J)

Fraction

Decimal

Percentage

WE DO with your teacher: copy every step into the space

Example 2 Give each answer in the form asked. Read each question twice.

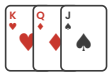
a As a decimal

One coin falls out of the money box. P(gold coin)



b Cards

One card from a 52-card deck. P(a red picture card)



c As a percentage

A die is rolled. P(a number greater than 4)



d Notes

Elena has two $5, four $10 and three $20 notes. She takes one at random. P($20 note)

YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. 6 QUESTIONS

REMEMBER P(E) = favourable ÷ total · if the total changes, the probability changes · never count an outcome twice

1 An Esky has 8 lemonade, 5 orange and 2 lime cans. How many cola cans must be added so that P(cola) = $\frac{2}{5}$?

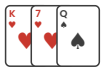


Answer

2 Start again with the Esky in Q1. How many cola cans must be added so that P(orange) = 0.25?

Answer

3 One card from a standard deck. Find P(a heart or a picture card). Careful: don't count any card twice.



Answer

4 One tile is chosen at random. Find P(an L or an M).



Answer

5 The spinner's angles are 180° red, 90° blue, 60° green and 30° yellow. Find P(green) and P(yellow).



Answer

6 A bag holds 40 marbles and P(red) = 0.35. How many red marbles are there? How many are not red?

Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. 3 PROBLEMS

1 The digits 1, 3, 5 and 8 are each used once to make a 4-digit number at random. Find P(it starts with 5), P(it is odd) and P(it is more than 5000).

Answer

2 A jar holds 60 lollies. P(red) = $\frac{1}{4}$ and P(green) = $\frac{1}{3}$; the rest are yellow. How many yellow lollies are there? Find P(yellow).

Answer

3 Kai picks a whole number from 1 to 20 at random. Kai wins on a multiple of 3 or a multiple of 4. Find P(Kai wins).

Answer

3 Complementary events

Probability • MA4-PRO-C-01 **MEDIUM ••**

LEARNING INTENTION

We are learning to identify the complement of an event, and use $P(\text{not E}) = 1 - P(E)$ to solve problems.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- ▸ *I can* write the complement of an event. ○○○
- ▸ *I can* show that $P(E) + P(\text{not E}) = 1$. ○○○
- ▸ *I can* use $P(\text{not E}) = 1 - P(E)$ with fractions, decimals and percentages. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Write each complement, then find the probabilities.

a Complement

Write the complement of rolling a 6 on a die. List its outcomes.



b Cards

A pack has 10 red, 6 yellow and 4 green cards. One is chosen. Find P(yellow) and P(not yellow).

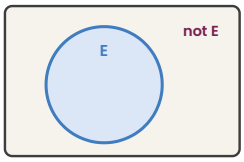
c Decimal

$P(\text{rain}) = 0.15$. Find P(no rain).



d Diagram

Die: E = an even number. Fill in the diagram, then find P(not E).



✎ YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER the complement must include every other outcome · $P(E) + P(\text{not E}) = 1$

ROUND 1 Questions 1–4: write the complement of each event. It must contain every other outcome.

1. Winning a hockey game

   Complement _____

2. Being born in winter

   Complement _____

3. Being under 15 years old

   Complement _____

4. Choosing a hearts card from a deck

   Complement _____

ROUND 2 Questions 5–8: a die is rolled. Find P(E) and P(not E).

5. E = at least 2

   P(E) _____

   P(not E) _____

6. E = at most 4

   P(E) _____

   P(not E) _____

7. E = a multiple of 3

   P(E) _____

   P(not E) _____

8. E = a number greater than 6

   P(E) _____

   P(not E) _____

3 Complementary events

Probability • MA4-PRO-C-01

HARD •••

YOU DO on your own: keep going, one round at a time. Show your working. **12 QUESTIONS**

REMEMBER $P(\text{not E}) = 1 - P(\text{E})$ · all probabilities add to 1 · 'not A or B' means neither A nor B

ROUND 3 Questions 9–11: write the probability of the complementary event.

9 $P(\text{choosing a jack from a deck}) = \frac{1}{13}$

Answer

10 $P(\text{shooting a goal}) = 61\%$

Answer

11 $P(\text{winning a prize}) = 0.15$

Answer

ROUND 4 Questions 12–14: a jar has 40 red, 25 blue, 50 black and 15 white jelly beans. Find each probability in simplest form.

12 Find $P(\text{not yellow})$.

Working

Answer

13 Find $P(\text{not red})$.

Working

Answer

14 Find $P(\text{not black or white})$.

Working

Answer

ROUND 5 Questions 15–17: the probabilities of all the outcomes add to 1. Find the missing probability.

15 Tomorrow: $P(\text{sunny}) = 0.45$, $P(\text{cloudy}) = 0.3$, $P(\text{rain}) = ?$

Working

Answer

16 A spinner is red, blue or green. $P(\text{red}) = \frac{1}{3}$, $P(\text{blue}) = \frac{1}{4}$. $P(\text{green}) = ?$

Working

Answer

17 A netball game: $P(\text{win}) = 55\%$, $P(\text{draw}) = 5\%$. $P(\text{lose}) = ?$

Working

Answer

ROUND 6 Questions 18–20: one tile is chosen at random from P R O B A B I L I T Y. Find each probability.

P	R	O	B	A	B	I	L	I	T	Y

18 Find $P(\text{not P})$.

Working

Answer

19 Find $P(\text{not a vowel})$.

Working

Answer

20 Find $P(\text{not A or B})$.

Working

Answer

WE DO with your teacher: copy every step into the space

Example 2 Use P(not E) = 1 - P(E). Give each answer in the form asked.

a Percentage

A golfer sinks 74% of putts. Find P(missing a putt).

b Missing value

A bag of toy cars holds only red, blue and white cars. P(red) = 0.4 and P(white) = 0.25. Find P(blue).

c Choose

The complement of 'winning a race' is: A coming last B coming second C not winning D coming 2nd or 3rd

d Raffle

Tahnee buys 5 of the 1000 raffle tickets. Find P(Tahnee does not win).



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. 6 QUESTIONS

REMEMBER P(E) + P(not E) = 1 · write an equation when one probability is a multiple of another

1. A car park has 450 cars, 100 motorbikes and 10 buses. P(not a car) = ?

   A $\frac{45}{56}$ B $\frac{110}{450}$ C $\frac{10}{560}$ D $\frac{11}{56}$

   Answer

2. A spinner is only red, blue or green. P(red) is twice P(blue), and P(green) = 0.1. Find P(red).

   Answer

3. P(not E) is three times P(E). Find P(E).

   Answer

4. A bag holds 50 counters. P(not red) = 0.64. How many red counters are there?

   Answer

5. A 2-digit number (10 to 99) is chosen at random. Find P(not a multiple of 10).

   Answer

6. Is P(E) = 0.7 with P(not E) = 0.4 possible? Explain.

   Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. 3 PROBLEMS

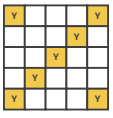
1. A bag of 40 toy cars holds only red, blue and white cars. P(red) = 0.4 and P(white) = 0.25. How many cars of each colour are there? What is P(a pink car)?

   Answer

2. A raffle sells 800 tickets for one prize. Jai buys some. P(Jai does not win) = 0.985. How many tickets did Jai buy?

   Answer

3. The grid game has 7 yellow squares out of 25. How many more squares must be coloured yellow so that P(not winning) = 0.6?



   Answer

4 Probability problems

Probability • MA4-PRO-C-01

MEDIUM ●●

LEARNING INTENTION

We are learning to choose and use the right probability skill to solve problems from lists, tables and words.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- ▸ *I can* decide which skill a problem needs. ○○○
- ▸ *I can* set out working with P(E) = favourable ÷ total. ○○○
- ▸ *I can* explain my answer in a sentence. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Use the checklist. Write the fraction in simplest form unless asked otherwise.

a A table

28 students on a trip to Japan were born in these countries. One is chosen at random. Find P(born in Asia).

Australia	17
Italy	3
Vietnam	4
Japan	1
China	3

b Letters

One tile is chosen. List the sample space. Is each letter equally likely?

S U C C E S S

c Percentage

Find P(an even number), as a percentage.



d Not

Alex takes one sock from 2 black, 2 blue and 2 red socks. Find P(not red).



✎ YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER P(E) = favourable ÷ total · simplest form · decimal: top ÷ bottom

ROUND 1 Questions 1–4: a ball is chosen at random from a barrel of balls numbered 1 to 100. Find each probability in simplest form.

1 Find P(at most 85).

85

Working _____

P = _____

2 Find P(a multiple of 7).

77

Working _____

P = _____

3 Find P(a two-digit number).

47

Working _____

P = _____

4 Find P(the number contains the digit 5).

55

Working _____

P = _____

ROUND 2 Questions 5–8: an 8-sided die has 2 red, 3 white, 1 blue and 2 yellow faces. Find each probability as a decimal.

5 Find P(blue).

B

Working _____

P = _____

6 Find P(yellow).

Y

Working _____

P = _____

7 Find P(not red).

R

Working _____

P = _____

8 Find P(white or yellow).

W

Working _____

P = _____

4 Probability problems

Probability • MA4-PRO-C-01

HARD •••

YOU DO on your own: keep going, one round at a time. Show your working. **12 QUESTIONS**

REMEMBER list outcomes carefully · count each object, not each name · 'or': never count an outcome twice

ROUND 3

Questions 9–11: a tennis tournament has 34 players from NSW, 20 from Victoria and 6 from Queensland. One is chosen at random.

9 Find P(from Queensland).

Working

Answer

10 Find P(not from Victoria).

Working

Answer

11 Find P(from Queensland or Victoria).

Working

Answer

ROUND 4

Questions 12–14: a whole number from 1 to 15 is chosen at random. List the outcomes, then find the probability.

12 Find P(an even number or a multiple of 5).

Outcomes { }

P =

13 Find P(not a factor of 12).

Outcomes { }

P =

14 Find P(a number that is prime and odd).

Outcomes { }

P =

ROUND 5

Questions 15–17: the 10 letters of KINGSCLIFF are written on cards. One card is chosen at random.

K	I	N	G	S	C	L	I	F	F

15 Is each letter equally likely? Explain.

Working

Answer

16 Find P(a letter that is in the word FISH).

Working

Answer

17 Find P(not a vowel).

Working

Answer

ROUND 6

Questions 18–20: a truck carries 240 boxes of hand sanitiser, 305 of soap, 335 of paper and 120 of pens. One box is chosen at random.

18 Find P(hand sanitiser).

Working

Answer

19 Find P(pens or soap).

Working

Answer

20 Find P(not paper).

Working

Answer

4 Probability problems

GOING FURTHER

WE DO with your teacher: copy every step into the space

Example 2 Harder problems: read twice, plan, then solve.

a Small chances
A charity raffle sells 100 000 tickets. Sophia buys 4. Find P(Sophia wins), as a decimal.



b Or
A whole number from 1 to 15 is chosen at random. Find P(a multiple of 3 or 5).



c How many?
Lollies: P(red) = 0.5, P(green) = 0.2 and the rest are purple. Find P(purple). How many of each colour in a bag of 10?

d No memory
A coin shows tails 7 times in a row. Find P(tails on the next toss).



✎ YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. — 6 QUESTIONS

REMEMBER plan before you calculate · a fair game gives every player the same probability of winning

1. Bag A has 10 lollies and bag B has 100. In both, P(red) = 0.5. Which bag gives the better chance of a red lolly? Explain.

Answer

2. A whole number from 1 to 50 is chosen at random. Find P(a multiple of 4 or a multiple of 6).

Answer

3. A 3-digit number from 100 to 999 is chosen at random. Find P(all three digits are the same).

Answer

4. A die is weighted so that P(6) = 0.25 and the other five numbers are equally likely. Find P(1).

Answer

5. In a class, P(a student is left-handed) = 0.15. There are 3 left-handed students. How many students are in the class?

Answer

6. Explain what is wrong: 'A game has two outcomes, win or lose, so P(win) = $\frac{1}{2}$.'

Answer

✎ WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. — 3 PROBLEMS

1. The canteen sold 120 lunches: 45 wraps, 30 sushi and the rest burgers. A receipt is chosen at random. Find P(burger). If 20 more burgers are sold, find the new P(burger).

Answer

2. Cards numbered 1 to 20: Ash wins on a multiple of 4, Bo on a prime, and Cy on any other number. Is the game fair? Who is most likely to win?

Answer

3. A bag holds 20 marbles and P(red) = 0.35. How many blue marbles must be added to make P(red) = 0.25?

Answer

5 Observed probability

Probability • MA4-PRO-C-01 **MEDIUM ••**

LEARNING INTENTION

We are learning to find observed probabilities from experiments and data, and compare them with theoretical probabilities.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* find an observed probability: frequency ÷ total trials. ○○○
- *I can* tell theoretical and observed probability apart. ○○○
- *I can* run a simulation with a random number generator. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Observed P(E) = number of times E happened ÷ total trials.

a Observed

A die is rolled 60 times and 2 comes up 15 times. Find the observed P(2).



b Compare

Find the theoretical P(2) for a fair die. Is the observed P(2) higher or lower?

c Spinner

Vinh spins 80 times: red 44, green 25, yellow 11. Find the observed P(red).



d Percentage

Sefina tosses a coin: 135 heads and 115 tails. Find the observed P(tails), as a %.



✎ YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER observed $P(E) = \frac{\text{number of times E happened}}{\text{total number of trials}}$ · theoretical: from equally likely outcomes · observed: from results

ROUND 1 Questions 1–4: find each observed probability as a decimal.

1 A coin is tossed 250 times: 135 heads. Find the observed P(tails).

Working _____
P = _____

2 A die is rolled 80 times: 1 or 6 comes up 25 times. Find the observed P(1 or 6).

Working _____
P = _____

3 A goalkeeper saves 18 of 24 penalties. Find the observed P(save).

Working _____
P = _____

4 73 of last year's 365 days had rain. Find the observed P(rain).

Working _____
P = _____

ROUND 2 Questions 5–8: theoretical (T) or observed (O)? Write T or O and give a reason.

5 40 of 100 Tweed residents surveyed own a dog, so P(owns a dog) = 0.4.

T or O _____
Reason _____

6 P(drawing an ace from a deck) = $\frac{1}{13}$.

T or O _____
Reason _____

7 60 of 200 cars passing the school were white, so P(white) = 0.3.

T or O _____
Reason _____

8 P(a random month starts with J) = $\frac{1}{4}$.

T or O _____
Reason _____

5 Observed probability

Probability • MA4-PRO-C-01

HARD •••

YOU DO on your own: keep going, one round at a time. Show your working. **11 QUESTIONS**

REMEMBER observed P = frequency ÷ total trials · 'at least 50' means 50 or more · a biased die has outcomes that are not equally likely

ROUND 3 Questions 9–11: the matches in 50 matchboxes were counted.

Matches	48	49	50	51	52	53
Boxes	3	6	20	16	4	1

9 Find the observed P(at least 50 matches), as a decimal.

Working _____

Answer _____

10 Find the observed P(not exactly 50 matches), as a decimal.

Working _____

Answer _____

11 The label says 'average contents 50'. Find the mean number of matches. Is the label fair? Ch 7

Working _____

Answer _____

ROUND 4 Questions 12–14: a die was rolled 80 times.

Outcome	1	2	3	4	5	6
Frequency	11	13	9	13	12	22

12 Find the observed P(1), as a percentage.

Working _____

Answer _____

13 Find the observed P(an even number), as a decimal.

Working _____

Answer _____

14 Is this die biased? Explain with numbers.

Working _____

Answer _____

ROUND 5 Questions 15–17: a pair of dice was rolled 50 times and the sum recorded.

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	0	2	4	6	5	5	9	6	8	3	2

15 Find the observed P(a composite sum), as a decimal.

Working _____

Answer _____

16 Which sum(s) had an observed probability of $\frac{3}{25}$?

Working _____

Answer _____

17 Which sum had an observed probability of 16%?

Working _____

Answer _____

ROUND 6 Questions 18–19: run a simulation and watch the observed probability change.

18 Use RanInt#(1,2) (1 = heads) to toss a coin 40 times, in blocks of 10. Record the heads so far and the observed P(heads) after each block.

Tosses	10	20	30	40
Heads so far				
Observed P(heads)				

19 What happens to the observed P(heads) as the tosses increase? Combine your results with 4 classmates (200 tosses) and comment.

Answer _____

WE DO with your teacher: copy every step into the space

Example 2 Read each table carefully: the total number of trials is the bottom of the fraction.

Matches in 50 boxes

Matches	48	49	50	51	52	53
Boxes	3	6	20	16	4	1

A die rolled 80 times

Outcome	1	2	3	4	5	6
Frequency	11	13	9	13	12	22

a Table
Matchboxes: find the observed P(exactly 51 matches).

b Fewer than
Matchboxes: find the observed P(fewer than 50 matches).

c Biased?
The die rolled 80 times: do you think it is biased? Give a reason.

d Simulate
Use RanInt#(1,6) to roll a die 20 times. Find your observed P(an even number).



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. **6 QUESTIONS**

REMEMBER frequency = observed P × number of trials · work backwards from a probability to a count

1. After 50 tosses, the observed P(heads) = 0.58. After 30 more tosses, the observed P(heads) for all 80 = 0.55. How many heads were in the last 30 tosses?

Answer

2. A spinner is spun 120 times and the observed P(red) = 0.35. How many times did it land on red?

Answer

3. A die is rolled 40 times: 1 five times, 2 eight times, 3 x times, 4 six times, 5 seven times and 6 nine times. Find x and the observed P(3).

Answer

4. After 20 games, the observed P(Kingscliff wins) = 0.65. They win the next 5 games. Find the new observed P(win).

Answer

5. Use the dice-sum table from Round 5. Find the mean sum. Ch 7

Answer

6. A biased coin lands tails 248 times in 400 tosses. Find the observed P(heads).

Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. **3 PROBLEMS**

1. Jo's family simulation gave 4 girls 4 times in 50 trials. The theoretical P(4 girls) = $\frac{1}{16}$ (you will see why in Stage 5). Jo runs 50 more trials and gets 4 girls twice. Find the observed probability for all 100 trials. Is it closer to $\frac{1}{16}$?

Answer

2. A spinner has 3 colours. In 200 spins: red 98, blue 52, green 50. Sketch what the spinner probably looks like, and explain.

Answer

3. Two classes test a coin. 8K gets 23 heads in 40 tosses; 8R gets 112 heads in 200 tosses. Find each observed P(heads). Which is the better estimate, and why?

Answer

6 Expected frequency

Probability • MA4-PRO-C-01

MEDIUM ●●

LEARNING INTENTION

We are learning to calculate the expected frequency of an event, and compare it with the observed frequency.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* find expected frequency = probability × number of trials. ○○○
- *I can* compare observed and expected frequencies. ○○○
- *I can* use an observed probability to make a prediction. ○○○

NOTES copy what your teacher writes on the board Date _____

WE DO with your teacher: copy every step into the space

Example 1 Expected frequency = probability × number of trials.

a Coin

A coin is tossed 50 times. Find the expected number of heads.



b Die

A die is rolled 72 times. Find the expected number of 4s.



c Spinner

Vinh spins 80 times. Find the expected number of reds. He got 44: compare.



d Rain

P(rain on an April day) = 30%. Find the expected number of rainy days in April.



YOU DO on your own: work down the box, one round at a time. Show your working. **8 QUESTIONS**

REMEMBER expected frequency = probability × number of trials, e.g. $\frac{1}{6} \times 60 = 10$

ROUND 1 Questions 1–4: find each expected frequency.

1 A die is rolled 300 times. Expected number of 6s?



P(E) _____
× trials _____
Expected _____

2 An 8-sided die has 2 red faces. It is rolled 200 times. Expected reds?



P(E) _____
× trials _____
Expected _____

3 A card is drawn from a deck and replaced, 260 times. Expected hearts?



P(E) _____
× trials _____
Expected _____

4 P(rain) = 0.35 each day. Expected rainy days in 20 days?



P(E) _____
× trials _____
Expected _____

ROUND 2 Questions 5–8: Vinh's spinner (½ red, ¼ green, ¼ yellow), 80 spins: red 44, green 25, yellow 11. Compare.

5 Red: expected and observed?

Working _____
Answer _____

6 Green: expected and observed?

Working _____
Answer _____

7 Yellow: expected and observed?

Working _____
Answer _____

8 Is the spinner fair? Explain.

Working _____
Answer _____

6 Expected frequency

Probability • MA4-PRO-C-01

HARD •••

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER expected = P × trials · for data, use observed P · small samples make poor predictions

ROUND 3 — Questions 9–11: a fair die is rolled 72 times. Find each expected frequency.

9 Expected number of 4s

Working _____

Answer _____

10 Expected number greater than 2

Working _____

Answer _____

11 Expected number of primes

Working _____

Answer _____

ROUND 4 — Questions 12–14: use the matchbox sample of 50 boxes to predict for 1000 boxes.

Matches	48	49	50	51	52	53
Boxes	3	6	20	16	4	1

12 Exactly 50 matches?

Working _____

Answer _____

13 At least 50 matches?

Working _____

Answer _____

14 Fewer than 49 matches?

Working _____

Answer _____

ROUND 5 — Questions 15–17: in 72 rolls of a die, 4 came up 9 times. Predict the number of 4s in 1000 rolls.

15 Using the theoretical probability (1 d.p.)

Working _____

Answer _____

16 Using the observed probability

Working _____

Answer _____

17 Which prediction do you trust more? Why?

Working _____

Answer _____

ROUND 6 — Questions 18–20: a pair of dice was rolled 50 times and the sum recorded.

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	0	2	4	6	5	5	9	6	8	3	2

18 Using the observed probability, how many sums of 8 are expected in 300 rolls?

Working _____

Answer _____

19 Using the observed probability, how many sums of 2 are expected? Why is this prediction unreliable?

Working _____

Answer _____

20 The theoretical P(sum of 7) = $\frac{1}{6}$ (Stage 5 shows why). Find the expected number of 7s in 50 rolls and compare.

Working _____

Answer _____

WE DO with your teacher: copy every step into the space

Example 2 Use theoretical probability when the outcomes are equally likely; otherwise use the observed probability.

a From data

Harper's die: 15 twos in 60 rolls. Using this, how many 2s are expected in 500 rolls?



b Matchboxes

20 of 50 sample boxes held exactly 50 matches. How many of 1000 boxes would you expect to?

c Not whole

A die is rolled 80 times. Find the expected number of 6s. It came up 22 times: comment.

d Lollies

P(red) = 0.5, P(green) = 0.2, P(purple) = 0.3. How many of each colour would you expect in a bag of 100?

YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. **6 QUESTIONS**

REMEMBER work backwards: P = expected ÷ trials · trials = expected ÷ P

1. The expected frequency of an event in 400 trials is 150. Find its probability.

Answer

2. A spinner is spun 250 times. The expected number of blues is 100. What fraction of the spinner is blue?

Answer

3. The expected number of white cars among 80 cars is 24. Find P(a car is not white).

Answer

4. A die is rolled n times and the expected number of 6s is 35. Find n.

Answer

5. A 4-colour spinner (equal sectors) gave red 52, blue 49, green 47, yellow 52 in 200 spins. Is it fair? Use expected frequencies.

Answer

6. Lollies: P(red) = 0.5, P(green) = 0.2 and the rest are purple. A bag has 9 purple lollies. How many lollies would you expect in the bag?

Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. **3 PROBLEMS**

1. A biased coin landed tails 248 times in 400 tosses. Predict the number of heads in 1500 tosses.

Answer

2. 70 of the canteen's last 200 orders were wraps. Next week it expects 900 orders. How many wraps should it make? Why make a few extra?

Answer

3. Roll a die: Dani wins on 1 or 2, Van on 3, 4 or 5, and Kai on 6. They play 60 games. Find the expected number of wins for each. Is the game fair?

Answer

E1 Venn diagrams

Extension • Stage 5 Path • MA5-PRO-P-01

MEDIUM ●●

LEARNING INTENTION

We are learning to read and draw Venn diagrams, and use them to find probabilities with 'and', 'or' and 'not'.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* read the four regions of a two-circle Venn diagram. ○○○
- *I can* draw a Venn diagram, starting with the overlap. ○○○
- *I can* tell inclusive 'or' from exclusive 'or' and find probabilities. ○○○

KEY IDEAS read these carefully, then highlight the key words

SELF-STUDY

KEY TERMS

Venn diagram: a rectangle (everyone) with a circle for each group

A and B: in both groups: the overlap

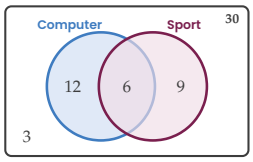
A or B (inclusive or): in A, in B, or in both

A or B but not both (exclusive or): A only or B only

neither: outside both circles

mutually exclusive: cannot happen together, so the circles do not overlap

DIAGRAM



12 computer only, 6 both, 9 sport only, 3 neither

STEPS

1. start with the overlap (both)
2. A only = A total − both
3. B only = B total − both
4. neither = total − the other three
5. P = number in the region(s) ÷ total

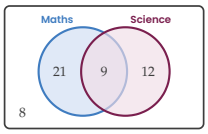
30 students: 18 computer, 15 sport, 6 both

WORKED worked examples: read each step, then cover it and try it yourself

Example 1 Read and draw Venn diagrams. Start with the overlap.

a Read

How many students were surveyed? How many like both subjects?



Add all four regions:
21 + 9 + 12 + 8 = **50** students

Both = the overlap = **9**

b Or

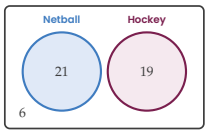
Same diagram. How many like Maths or Science? How many like Maths or Science, but not both?

Or = everything inside the circles:
21 + 9 + 12 = **42**

Not both = only the outer parts: 21 + 12 = **33**

c Mutually exclusive

Are netball and hockey mutually exclusive? How many play netball or hockey?

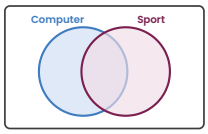


The circles do not overlap: no-one plays both, so **yes**, they are mutually exclusive.

Netball or hockey: 21 + 19 = **40**

d Draw

30 students: 18 play computer games, 15 play sport, 6 do both. Complete the diagram.

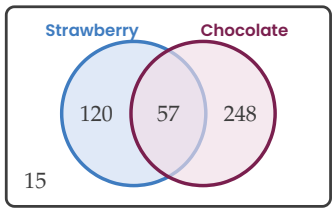


Overlap: **6**. Computer only: 18 − 6 = **12**. Sport only: 15 − 6 = **9**. Neither: 30 − 27 = **3**.

✎ YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER A and B: the overlap · A or B: inside either circle · not both: the two outer parts only · neither: outside the circles · Venn diagrams are not drawn to scale

ROUND 1 Questions 1–4: use this Venn diagram of an ice-cream survey.



Each number is a count of people.

1 How many people were surveyed?

Answer _____

2 How many liked strawberry or chocolate, but not both?

Answer _____

3 How many liked neither flavour?

Answer _____

4 Find P(a person likes both flavours).

Answer _____

ROUND 2 Questions 5–8: are the two events mutually exclusive or overlapping? Explain.

5 'surfs' and 'is left-handed'

ME / overlap _____

Reason _____

6 'born in Australia' and 'born in Japan'

ME / overlap _____

Reason _____

7 Die: 'an even number' and 'a 3'

ME / overlap _____

Reason _____

8 Die: 'an even number' and 'greater than 3'

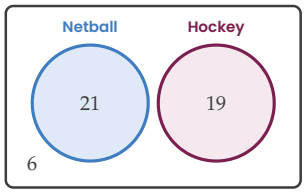
ME / overlap _____

Reason _____

✎ YOU DO on your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER start with the overlap · A only = A − both · neither = total − (A only + both + B only) · P = region ÷ total

ROUND 3 Questions 9–11: this Venn diagram shows the sports some students play.



9 How many students are in the diagram?

Answer _____

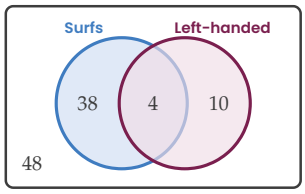
10 How many play netball or hockey?

Answer _____

11 How many play netball and hockey?

Answer _____

ROUND 4 Questions 12–14: one person from this diagram is chosen at random. Give each probability as a decimal.



12 Find P(surfs).

Answer _____

13 Find P(surfs or is left-handed).

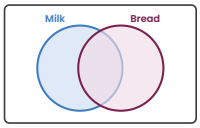
Answer _____

14 Find P(left-handed and does not surf).

Answer _____

ROUND 5 Questions 15–17: 30 customers: 20 bought milk, 13 bought bread and 1 bought neither. Draw a Venn diagram first.

15 How many bought both milk and bread?



Answer _____

16 How many bought milk only?

Answer _____

17 Find P(bread only).

Answer _____

ROUND 6 Questions 18–20: a group has 32 students with blue eyes, 38 with brown eyes and 10 with neither.

18 Are 'blue eyes' and 'brown eyes' mutually exclusive here? Why?

Answer _____

19 How many students are in the group?

Answer _____

20 Find P(not brown eyes), as a percentage.

Answer _____

WORKED worked examples: read each step, then cover it and try it yourself

Example 2 Find probabilities from Venn diagrams: the total of everyone goes on the bottom.

a And / not

Use Example 1d. Find P(plays sport but not computer games).

Sport but not computer = sport only = 9.

$P = \frac{9}{30} = \frac{3}{10}$

b Exclusive or

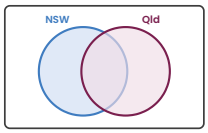
Use Example 1d. Find P(plays sport or computer games, but not both).

Not both = computer only + sport only = 12 + 9 = 21.

$P = \frac{21}{30} = \frac{7}{10}$

c Draw

130 people: 44 like NSW holidays, 66 like Queensland, 20 like both. Complete the diagram.



Overlap **20**. NSW only 44 − 20 = **24**. Qld only 66 − 20 = **46**. Neither 130 − 90 = **40**.

d Probability

Use part c. Find P(likes NSW only) and P(likes neither).

$P(\text{NSW only}) = \frac{24}{130} = \frac{12}{65}$

$P(\text{neither}) = \frac{40}{130} = \frac{4}{13}$

YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. **6 QUESTIONS**

REMEMBER P(A or B) = P(A) + P(B) − P(A and B), or draw the diagram and add the regions

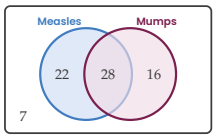
1 All 30 students in a class study IT or Chemistry (or both). 7 study both and 20 study IT. How many study Chemistry? Find P(Chemistry).

Answer

2 40 students at camp: 35 chose kayaking, 20 chose bushwalking, and everyone chose at least one. How many chose both? Find P(both) and P(not bushwalking).

Answer

3 The diagram shows children who have had measles or mumps. Find P(measles), P(measles or mumps) and P(neither).



Answer

4 P(A) = 0.6, P(B) = 0.5 and P(A and B) = 0.3. Draw a Venn diagram for 100 people and find P(A or B).

Answer

5 A Venn diagram has A only = x, both = 5, B only = 2x and neither = 4. The total is 39. Find x and P(A).

Answer

6 Can A and B be mutually exclusive if P(A) = 0.7 and P(B) = 0.5? Explain.

Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. **3 PROBLEMS**

1 120 Year 8 students: 70 have a dog, 45 have a cat and 25 have neither. Draw a Venn diagram. Find P(dog and cat) and P(exactly one of the two).

Answer

2 Kingscliff Nippers has 48 members: 30 swim, 26 do board and 5 do neither. Find P(swims but does not do board).

Answer

3 A number from 1 to 20 is chosen. A = even, B = a multiple of 3. Put the numbers in a Venn diagram. Find P(A or B) and P(A or B but not both).

Answer

E2 Two-way tables

Extension • Stage 5 Path • MA5-PRO-P-01

MEDIUM ●●

LEARNING INTENTION

We are learning to read, complete and draw two-way tables, convert between Venn diagrams and two-way tables, and find probabilities.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- ▸ *I can* complete a two-way table using the totals. ○○○
- ▸ *I can* convert between a Venn diagram and a two-way table. ○○○
- ▸ *I can* find probabilities with 'and', 'or' and 'not' from a table. ○○○

KEY IDEAS

read these carefully, then highlight the key words

SELF-STUDY

KEY TERMS

two-way table: rows for one attribute, columns for another

each **cell** is an 'and', e.g. Year 7 *and* can swim

totals: row totals, column totals and the **grand total** (bottom right)

missing cell = total − the known cells

VENN DIAGRAM ↔ TABLE

	B	not B	Total
A	both	A only	A total
not A	B only	neither	
Total	B total		grand

overlap → 'A and B' cell; outside both circles → 'not A and not B' cell

STEPS

1. fill in every total you can
2. find missing cells by subtracting
3. check: rows and columns add up
4. P = cell(s) ÷ grand total

'or': add the cells, but count each cell once

WORKED

worked examples: read each step, then cover it and try it yourself

Example 1 Read the two-way table. Each cell is an 'and'.

	Can swim	Cannot swim
Year 7	13	2
Year 8	9	3

a Total

How many students are in the table?

Add every cell:
13 + 2 + 9 + 3 = **27**

b Or

How many students are in Year 7 or cannot swim?

All of Year 7 (13 + 2) plus Year 8 students who cannot swim (3):
15 + 3 = **18**
Don't count the 2 twice.

c And

How many students are in Year 7 and cannot swim?

The cell in the Year 7 row and the Cannot swim column: **2**

d Probability

One student is chosen at random. Find P(a Year 8 student who can swim).

Year 8 and can swim: 9 of 27.
$P = \frac{9}{27} = \frac{1}{3}$

✎ YOU DO

on your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER each cell is an 'and' · 'or': add the cells, counting each once · P = cell ÷ grand total

ROUND 1 Questions 1–4: the players in a soccer club are grouped by age and weight.

	Heavy	Light
Junior	64	96
Senior	144	32

1. How many players does the club have?

Answer _____

2. How many players are juniors or light?

Answer _____

3. How many are juniors or light, but not both?

Answer _____

4. Find P(a player chosen at random is a senior).

Answer _____

ROUND 2 Questions 5–8: complete the table first. It shows which students need glasses.

	Needs glasses	Does not	Total
Year 9	22		82
Year 10		85	98
Total		145	

5. How many Year 9 students do not need glasses?

Answer _____

6. How many Year 10 students need glasses?

Answer _____

7. How many students are there in total?

Answer _____

8. Find P(a student chosen at random needs glasses).

Answer _____

YOU DO on your own: keep going, one round at a time. Show your working. **12 QUESTIONS**

REMEMBER missing cell = total − known cells · overlap → 'A and B' cell · outside both circles → 'not A and not B' cell

ROUND 3 Questions 9–11: complete this table about liking broccoli, then answer.

	Likes	Dislikes	Total
Year 7	60		150
Year 8		40	
Total	130		

9 How many students dislike broccoli?

Answer

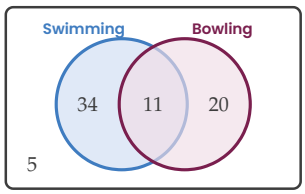
10 Find P(Year 7 or dislikes broccoli).

Answer

11 Find P(Year 8 and likes broccoli).

Answer

ROUND 4 Questions 12–14: convert the Venn diagram to a two-way table, then answer.



12 Complete the table.

	S	not S	Tot
B			
not B			
Tot			

13 Find P(a person swims).

Answer

14 Find P(bowls but does not swim).

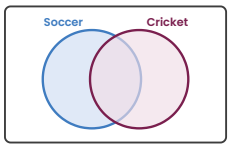
Answer

ROUND 5 Questions 15–17: convert the two-way table to a Venn diagram, then answer.

	Cricket	Not cricket
Soccer	28	15
Not soccer	20	7

TV sports watched by 70 people

15 Draw the Venn diagram.



16 Find P(watches soccer or cricket, but not both).

Answer

17 Find P(does not watch cricket).

Answer

ROUND 6 Questions 18–20: a sports high school surveyed students about hockey and judo. Complete the table, then answer.

	Judo	Not judo	Total
Hockey	128		360
Not hockey	100	40	
Total			

18 Find P(does judo).

Answer

19 Find P(does judo but not hockey).

Answer

20 Find P(hockey or judo, but not both).

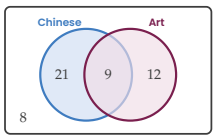
Answer

WORKED worked examples: read each step, then cover it and try it yourself

Example 2 Complete tables, use them for probabilities, and convert from a Venn diagram.

	Under 18	18 and over	Total
Saturday		142	198
Sunday	45		
Total			344

A cinema's audience on two days



a Complete
Complete the cinema table.

198 − 142 = **56**
344 − 198 = **146** (Sunday)
146 − 45 = **101**
56 + 45 = **101**; 142 + 101 = **243**

b And
One person is chosen at random. Find P(on Sunday and aged 18 or over).

Sunday and 18+: 101 of 344.
$P = \frac{101}{344}$

c Or
Find P(on Sunday or aged 18 or over).

All of Sunday (146) plus Saturday 18+ (142): 288.
$P = \frac{288}{344} = \frac{36}{43}$

d Convert
Show the Venn diagram (50 students) in a two-way table: Art / not Art down the side.

	Ch	not	Tot
Art	9	12	21
not	21	8	29
Tot	30	20	50

✎ YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. 6 QUESTIONS

REMEMBER draw the table or the diagram first · rows and columns must add to the totals

1 Cars at Rusty Motors. Show the Venn diagram in a two-way table. Find P(has GPS).



Answer

2 Use Question 1. Find P(4 doors and GPS) and P(no GPS).

Answer

3 A two-way table has grand total 80. The A total is 30, the B total is 45 and the A-and-B cell is 12. Find the 'not A and not B' cell.

Answer

4 8K has 30 students: 18 walk to school, and 9 of the walkers are in the band. 4 students who don't walk are in the band. Make a two-way table. Find P(in the band).

Answer

5 40% of Year 8 surf, 25% skate and 10% do both. Find P(neither), as a percentage.

Answer

6 P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2. Complete a two-way table of probabilities and find P(neither).

Answer

✎ WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. 3 PROBLEMS

1 The canteen had 200 orders: 120 included a drink, 90 were hot food, and 50 were hot food with a drink. Make a two-way table. Find P(hot food or a drink, but not both).

Answer

2 150 Tweed residents: 40 own a boat, 55 live on the river and 18 do both. Find P(owns a boat but does not live on the river) and P(neither).

Answer

3 Explain why a two-way table can show Years 7, 8 and 9 (down the side) more easily than a Venn diagram.

Answer

CHAPTER SUMMARY

the key facts from every lesson, and how confident you are with each

HOW CONFIDENT AM I?

○○○ not yet · getting there · got it

1	Chance and the sample space	○○○	2	Probability of an event	○○○
3	Complementary events	○○○	4	Probability problems	○○○
5	Observed probability	○○○	6	Expected frequency	○○○
E1	Venn diagrams	○○○	E2	Two-way tables	○○○

Key facts

Sample space: every possible outcome, each written once, in braces { } (L1)

P(E) $= \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$ when outcomes are equally likely (L2)

'or' for different groups: add the counts, e.g. P(red or blue) (L2)

Checklist: total → favourable → 'not'? → form asked → check (L4)

More trials: the observed probability usually gets closer to the theoretical one (L5)

Fair or biased? Close to expected: probably fair. Far off over many trials: probably biased (L6)

Two-way table: each cell is an 'and'; missing cell = total − known cells; P = cell ÷ grand total (E2)

Probability scale: 0 impossible · $\frac{1}{2}$ even chance · 1 certain (L1)

Converting: fraction → decimal: top ÷ bottom; decimal → percentage: × 100% (L2)

Complement: P(not E) = 1 − P(E); all probabilities add to 1 (L3)

Observed P(E) $= \frac{\text{number of times E happened}}{\text{total number of trials}}$ (L5)

Expected frequency = probability × number of trials (L6)

Venn diagram: A and B = the overlap; A or B = inside either circle; not both = the outer parts; mutually exclusive = no overlap (E1)

IF YOU WERE AWAY

two examples from each lesson, fully worked: read them, then try the lesson's first round

Lesson 1 • Chance and the sample space

the lesson is on p. 2

Example 1: a sample space

List the sample space for a spinner with sectors red, blue, red, green.

Step 1 Write each colour once: {red, blue, green}.

Step 2 Count: 3 outcomes.

Step 3 Red has 2 of the 4 sectors, so the outcomes are **not equally likely**.

Example 2: chance words

The forecast says a 30% chance of rain. Describe the chance.

Step 1 Compare with 50% (even chance): 30% is less.

Step 2 So rain is **unlikely**.

Step 3 It is not 0%, so it is still possible.

Lesson 2 • Probability of an event

the lesson is on p. 5

Example 1: P(E) from counters

A bag has 5 red, 3 blue and 4 green counters. Find P(red).

Step 1 Total = 5 + 3 + 4 = 12.

Step 2 Favourable: 5 red.

Step 3 P(red) $= \frac{5}{12}$

Example 2: a percentage

A die is rolled. Find P(a number greater than 4), as a percentage.

Step 1 Favourable: {5, 6}, 2 of 6.

Step 2 $\frac{2}{6} = \frac{1}{3}$

Step 3 $\frac{1}{3} \times 100\% = 33\frac{1}{3}\% \approx 33.3\%$

Lesson 3 • Complementary events

the lesson is on p. 8

Example 1: P(not E) with decimals

$P(\text{rain}) = 0.15$. Find $P(\text{no rain})$.

Step 1 'No rain' is the complement of 'rain'.

Step 2 $\text{P}(\text{no rain}) = 1 - 0.15 = \mathbf{0.85}$

Example 2: P(not E) with fractions

A pack has 10 red, 6 yellow and 4 green cards. Find $P(\text{not yellow})$.

Step 1 $\text{P}(\text{yellow}) = \frac{6}{20} = \frac{3}{10}$

Step 2 $\text{P}(\text{not yellow}) = 1 - \frac{3}{10} = \mathbf{\frac{7}{10}}$

Lesson 4 • Probability problems

the lesson is on p. 11

Example 1: an 'or' problem

A number from 1 to 15 is chosen. Find $P(\text{a multiple of 3 or 5})$.

Step 1 List: $\{3, 5, 6, 9, 10, 12, 15\}$ (15 once).

Step 2 7 of 15: $\text{P} = \mathbf{\frac{7}{15}}$

Example 2: no memory

A coin has shown tails 7 times in a row. Find $P(\text{tails next})$.

Each toss is a fresh start: the coin does not remember.

$\text{P}(\text{tails}) = \mathbf{\frac{1}{2}}$

Lesson 5 • Observed probability

the lesson is on p. 14

Example 1: observed probability

A die is rolled 60 times; 2 comes up 15 times. Find the observed $P(2)$.

Step 1 Observed P = times ÷ trials $= \frac{15}{60} = \mathbf{\frac{1}{4}}$

Step 2 Theoretical $\text{P}(2) = \frac{1}{6}$, so 2 came up more than expected.

Example 2: from a table

Matches in 50 boxes: 48 (3), 49 (6), 50 (20), 51 (16), 52 (4), 53 (1). Find the observed $P(\text{fewer than } 50)$.

Step 1 Fewer than 50: 48 or 49, $3 + 6 = 9$ boxes.

Step 2 $\text{P} = \mathbf{\frac{9}{50}}$

Lesson 6 • Expected frequency

the lesson is on p. 17

Example 1: expected frequency

A die is rolled 72 times. Find the expected number of 4s.

Expected = P × trials $= \frac{1}{6} \times 72 = \mathbf{12}$

Example 2: predicting from data

Harper's die gave 15 twos in 60 rolls. How many 2s in 500 rolls?

Step 1 Observed $\text{P}(2) = \frac{15}{60} = \frac{1}{4}$

Step 2 $\frac{1}{4} \times 500 = \mathbf{125}$

Lesson E1 • Venn diagrams

the lesson is on p. 20

Example 1: drawing a Venn diagram

30 students: 18 computer, 15 sport, 6 both.

Step 1 Overlap: 6.

Step 2 Computer only $18 - 6 = 12$; sport only $15 - 6 = 9$.

Step 3 Neither: $30 - 27 = 3$.

Example 2: exclusive or

Same diagram: find P(sport or computer, but not both).

Not both = the two outer parts: $12 + 9 = 21$.

$\mathrm{P} = \frac{21}{30} = \mathbf{\frac{7}{10}}$

Lesson E2 • Two-way tables

the lesson is on p. 23

Example 1: reading a two-way table

Year 7: 13 can swim, 2 cannot; Year 8: 9 can, 3 cannot. Find P(Year 8 and can swim).

Step 1 Grand total 27.

Step 2 The Year 8 / can swim cell: 9.

Step 3 $\mathrm{P} = \frac{9}{27} = \mathbf{\frac{1}{3}}$

Example 2: a missing cell

Saturday: under 18 ?, 18 and over 142, total 198. Find the missing cell.

missing = total − known = $198 - 142 = \mathbf{56}$

R1 [L1] A die is rolled. Order from least to most likely: A a 1, B an odd number, C at most 5.

R2 [L2] An Esky has 6 lemonade and 4 lime cans. How many colas must be added so that $P(\text{cola}) = \frac{1}{3}$?

R3 [L2] One card from a deck. Find P(a red card or a king).

R4 [L2] A spinner has angles 120° red, 150° blue and 90° green. Find P(blue).

R5 [L3] $P(\text{not E}) = 0.62$. Find P(E).

R6 [L3] A spinner is red, green or blue. $P(\text{red}) = 2 \times P(\text{green})$ and $P(\text{blue}) = 0.4$. Find P(green).

R7 [L3] A 3-digit number (100 to 999) is chosen at random. Find P(not a multiple of 5).

R8 [L4] A whole number from 1 to 30 is chosen. Find P(a multiple of 3 or 5).

R9 [L4] A die is weighted so $P(6) = 0.4$ and the other numbers are equally likely. Find P(2).

R10 [L5] After 40 games, the observed $P(\text{win}) = 0.6$. The team then loses 10 games in a row. Find the new observed P(win).

R11 [L5] A die is rolled 120 times: 1 (18), 2 (22), 3 (20), 4 (19), 5 (21), 6 (x). Find x and the observed P(6).

R12 [L6] The expected frequency of an event in 400 trials is 140. Find its probability.

R13 [L6] The observed P(a car is white) = 0.3. Predict the number of white cars among 450 cars.

R14 [L6] A 4-sector spinner (equal sectors) gave 160 reds in 400 spins. Is it fair? Use the expected frequency.

R15 [E1] 60 students: 35 play sport, 28 play an instrument and 12 do both. How many do neither? Find P(exactly one).

R16 [E1] On a die, are 'a prime number' and 'an even number' mutually exclusive? Explain.

R17 [E1] P(A) = 0.45, P(B) = 0.3 and P(A and B) = 0.1. Find P(neither).

R18 [E2] Year 7: 40 like, total 70. Year 8: 25 dislike. Grand total 130. Find P(Year 8 and likes).

R19 [E2] A Venn diagram has A only 14, both 6, B only 10 and neither 20. Make a two-way table and find P(not A).

R20 [E2] Use the previous question. Find P(A or B, but not both).

EXIT TICKET Lesson 1: Chance and the sample space Name _____ Class _____

Medium ●● List the sample space for the number of heads when 2 coins are tossed.

Hard ●●● Are those outcomes equally likely? Explain.

Going further Design a 6-sector spinner where red is likely and blue is unlikely but possible.

EXIT TICKET Lesson 2: Probability of an event Name _____ Class _____

Medium ●● One card from a deck. Find P(a picture card).

Hard ●●● A 10-sided die (1 to 10). Find P(a multiple of 4), as a decimal.

Going further A bag has 4 red counters and some blue. P(red) = 0.2. How many blue?

EXIT TICKET Lesson 3: Complementary events Name _____ Class _____

Medium ●● Write the complement of 'at least 3' on a die, and its probability.

Hard ●●● P(red) = 0.3, P(blue) = 0.45, the rest green. Find P(not green).

Going further P(not E) is four times P(E). Find P(E).

EXIT TICKET Lesson 4: Probability problems Name _____ Class _____

Medium ●● A number from 1 to 100 is chosen. Find P(a perfect square).

Hard ●●● A number from 1 to 20 is chosen. Find P(a multiple of 2 or 5).

Going further Roll a die: A wins on 1 or 2, B on 3 to 6. Is it fair? Explain.

✂

SUMMARISE LESSON 1 IN YOUR OWN WORDS

Chance and the sample space • the back of your exit ticket

✂

SUMMARISE LESSON 2 IN YOUR OWN WORDS

Probability of an event • the back of your exit ticket

✂

SUMMARISE LESSON 3 IN YOUR OWN WORDS

Complementary events • the back of your exit ticket

✂

SUMMARISE LESSON 4 IN YOUR OWN WORDS

Probability problems • the back of your exit ticket

✂

EXIT TICKETS cut along the dashed lines • write your summary on the back • hand it to your teacher

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EXIT TICKET Lesson 5: Observed probability Name _____ Class _____

Medium ●● 400 tosses gave 150 tails. Find the observed P(tails), as a decimal.

Hard ●●● After 20 shots, observed P(goal) = 0.45. How many goals?

Going further The player then scores 5 of the next 5 shots. Find the new observed P(goal).

EXIT TICKET Lesson 6: Expected frequency Name _____ Class _____

Medium ●● A spinner is ¼ red. Find the expected number of reds in 200 spins.

Hard ●●● The expected frequency is 45 in 180 trials. Find the probability.

Going further Observed P = 0.35 after 60 trials. Predict the number in 500 trials.

EXIT TICKET Lesson E1: Venn diagrams Name _____ Class _____

Medium ●● Venn: A only 8, both 3, B only 5, neither 4. Find P(A or B).

Hard ●●● Same diagram: find P(A or B, but not both).

Going further 40 students: 22 surf, 15 skate and 7 do neither. How many do both?

EXIT TICKET Lesson E2: Two-way tables Name _____ Class _____

Medium ●● Year 7: 12 like, 8 dislike. Year 8: 15 like, 5 dislike. Find P(Year 7 and likes).

Hard ●●● Same table: find P(Year 8 or dislikes).

Going further Draw the Venn diagram for that table (circles: Year 7, likes).

✂

SUMMARISE LESSON 5 IN YOUR OWN WORDS

Observed probability • the back of your exit ticket

✂

SUMMARISE LESSON 6 IN YOUR OWN WORDS

Expected frequency • the back of your exit ticket

✂

SUMMARISE LESSON E1 IN YOUR OWN WORDS

Venn diagrams • the back of your exit ticket

✂

SUMMARISE LESSON E2 IN YOUR OWN WORDS

Two-way tables • the back of your exit ticket

✂

HOMEWORK STRIPS — after each lesson, cut out its strip (start at the top right), do it at home, and hand it in next lesson

HW 7 E1 Venn ✂

1 Venn: A only 9, both 4, B only 7, neither 5. Find P(A).

2 Same diagram: find P(A or B).

3 Same diagram: find P(exactly one of A, B).

HW 4 L4 Problems ✂

1 A number from 1 to 30. Find P(a multiple of 4 or 6).

2 P(left-handed) = 0.12 and 6 students are left-handed. How many students?

3 Roll a die: Ann wins on a prime, Bo on any other number. Is it fair?

HW 1 L1 Chance ✂

1 List the sample space for the number of heads when 3 coins are tossed.

2 Are those outcomes equally likely?

3 Order from least to most likely: 0.3, $\frac{2}{5}$, 35%.

HW 8 E2 Tables ✂

1 Table: A and B 6, A not B 9, B not A 5, neither 10. Find P(B).

2 Same table: find P(A or B).

3 Same table: find P(not A and not B).

HW 5 L5 Observed ✂

1 250 tosses gave 140 heads. Find the observed P(heads).

2 After 30 spins, observed P(red) = 0.4. The next 10 are all red. Find the new observed P(red).

3 Why do two classes get different observed probabilities?

HW 2 L2 P(E) ✂

1 One card from a deck. Find P(a red picture card).

2 An Esky has 6 lemonade and 4 lime. How many colas must be added so that P(lemonade) = $\frac{1}{2}$?

3 A spinner has a 90° red sector. Find P(red).

HW 9 Before the test ✂

1 **L3** One tile from KINGSCLIFF. Find P(not a vowel).

2 **L6** Find the expected number of 4s in 150 rolls.

3 **E1** On a die, are ‘a multiple of 3’ and ‘a 5’ mutually exclusive?

HW 6 L6 Expected ✂

1 The expected frequency is 90 in 360 trials. Find P(E).

2 A die is rolled n times; the expected number of 6s is 42. Find n.

3 Observed P = 0.15. Predict the number in 2000 trials.

HW 3 L3 Complements ✂

1 P(not E) = 0.35. Find P(E).

2 P(red) = 0.25, P(blue) = 0.4; the rest is green. Find P(green).

3 A number from 1 to 50. Find P(not a multiple of 7).

HW 1 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 2 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 3 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 4 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 5 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 6 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 7 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 8 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working:

HW 9 Name and working

Name _____

Class _____ Due _____

How did it go? ○○○ easy · okay · stuck

More working: